

# MUHAMMAD USMAN

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[LinkedIn](#) | [Portfolio](#)

**PROFILE:** Mechanical Engineering graduate with expertise in CAD design (SolidWorks), mechanical systems simulation and hands-on maintenance experience at FFC. Passionate about robotics, manufacturing, and problem-solving.

## EDUCATION

- **Bachelor of Mechanical Engineering** | PIEAS, Islamabad, Pakistan | **CGPA: 3.74** | 2021–2025  
**Relevant Coursework:** Statistics & Dynamics, Refrigeration, Power Plants, Engineering Design, Mechanics of Materials, Thermodynamics, Fluid Mechanics, Manufacturing Processes, Heat and Mass Transfer,
- **Intermediate** (F.Sc. Pre-Engineering) | Punjab College of Technology, Sargodha | 99.8% | 2019–2021
- **Matriculation** | Government High School Verowal, Sargodha | 96.3% | 2017–2019

## SKILLS

**Technical skills:** SolidWorks, Microsoft office, Craftware, Proteous, ANSYS, Linkages, CoppeliaSim, Hands-on machining (milling, drilling, grinding, tapping, welding).

**Soft skills:** Communication, Public Speaking, Teamwork, Leadership, Multilingual (Punjabi, Urdu, English), Project coordination and documentation, Problem Solving.

## PROFESSIONAL EXPERIENCE

- FAUJI FERTILIZER COMPANY LIMITED-GOTH MACHI PLANT** Rahimyar Khan, Punjab, Pakistan  
Maintenance Engineer Intern June 2024 – August 2024
- Supported mechanical maintenance activities for static and rotary equipment, including valves, heat exchangers, piping, cooling towers, and boilers.
  - Gained exposure to valve restoration processes, gas and plasma cutting techniques and machine shop operations.

- FINAL YEAR PROJECT HANDS-ON MANUFACTURING EXPERIENCE** PIEAS University, Islamabad
- Performed milling, drilling, grinding, tapping, and welding for robot components. October 2024- June 2025
  - Operated workshop machines and tools to fabricate precise metal parts.
  - Interpreted engineering drawings and CAD models for manufacturing.
  - Contributed to assembly and successful prototyping of the robot.

## ACADEMIC PROJECTS

- DESIGN AND DEVELOPMENT OF TRACK MOBILE ROBOT FOR FIRE RESCUE OPERATIONS (Final Year Project)** PIEAS University, Islamabad  
October 2024- June 2025

- Designed and developed a tracked robot to navigate hazardous terrains during fire rescue operations.
- Utilized **SolidWorks** for 3D CAD modeling, ensuring accuracy in manufacturing.
- Simulated robot operations using **CoppeliaSim** for dynamic testing and control validation.

- DESIGN OF SUGAR CANE MILL** PIEAS University, Islamabad  
CAD Design Project October 2023- November 2023

- Created a detailed 3D model of a sugar cane mill using **SolidWorks** for a semester project.
- Applied mechanical design principles and assembly techniques to model complex industrial equipment.

- BOX DETECTION USING YOLOV8** PIEAS University, Islamabad  
Computer Vision Project March 2025- May 2025

- Collected and annotated custom dataset of 1,000+ images using Roboflow.
- Trained YOLOv8 model on Google Colab GPU, achieving high detection accuracy for box identification tasks.
- Gained experience in machine learning pipelines, data handling, and model evaluation.

## ONLINE COURSES AND CERTIFICATES

- Utilities, Safety & Environmental Care in Oil & Gas Industry 2024
- Air Conditioning Equipment Selection, Design and Sizing 2025
- Basics of Air Conditioning & Heat Load Calculation 2025
- Introduction to Mechanical Engineering Design and Manufacturing with Fusion 360 2025

## AWARDS AND HONORS

- Awarded PEEF scholarship twice for academic excellence.
- Secured 1st position at Tehsil level in Sargodha Board Matriculation exams.
- Awarded fully funded scholarship during F.Sc. Pre-Engineering.
- Received Government of Pakistan Chartered Accountant scholarship worth PKR 1 million.
- Runner-up in PIEAS university badminton championship.
- Active member of ASME PIEAS.